



Shadow banking, financial regulation, and bank risk in China[☆]

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ABSTRACT

China's 2018 rules on wealth management products (WMPs) have led to determined efforts by regulators to reduce risk in the shadow banking system. Through theoretical model and empirical testing, we explore the impact of the regulation on WMPs and bank performance. Our study finds that after the regulation, banks changed the issuance structure and their profit and risk reduced. The capital did not flow back to on-balance sheet business and the return of WMPs decreased after the regulation.

1. Introduction

Shadow banking can be understood as off-balance-sheet financing that banks use for economizing their regulatory capital (Huang, 2018), a phenomenon that has emerged in the United States and Europe over two decades (Hodula, 2022). In the U.S., shadow banking evolved from traditional commercial activities of loan and deposit issuance towards a securitized banking business model (Meeks et al., 2017). In this process, banks, driven by regulatory arbitrage motivations, securitize higher-risk underlying assets, which lead to the accumulation of risk in the financial system. After the 2007 financial crisis, a similar trend emerged in China, where the scale of shadow banking rapidly expanded, and it exacerbated systemic risk and financial instability (Zhao and Yao, 2024). In China, shadow banking is mainly defined as off-balance-sheet activities that exist outside the formal banking system, such as WMPs (Chen et al., 2021). Through WMPs, banks can invest capitals in sectors with high risk while bypassing financial regulations, such as capital requirements, loss provision or deposit reserve.

Compared to asset-backed securitization, WMPs possess some unique characteristics. Firstly, by offering higher returns than deposits, WMPs compete with formal banking sector for funds from individuals and business sectors. WMPs usually contain higher risks but are sold by banks as low-risk products akin to deposits (Hachem and Song, 2016). Secondly, many WMPs prior to 2018 included guarantees on the principal by banks, which suggest that in case of default, banks would bear the loss. Thirdly, banks can expand WMPs without increasing regulatory capital, which makes WMPs more profitable than on-balance sheet loans, even when asset-side risks are similar (Luo et al., 2019).

In recent years, China's regulators have gradually realized the potential risks of shadow banking. As one of the main sources of funds for shadow banking, WMPs have become a key regulatory object (Chen et al., 2020). On April 27, 2018, China's central bank issued NAMR (New Asset Management Regulation). The core measures of the regulation include prohibiting banks from making commitments to investors regarding returns or principal and restricting the off-balance sheet leverage levels of banks. Similar to

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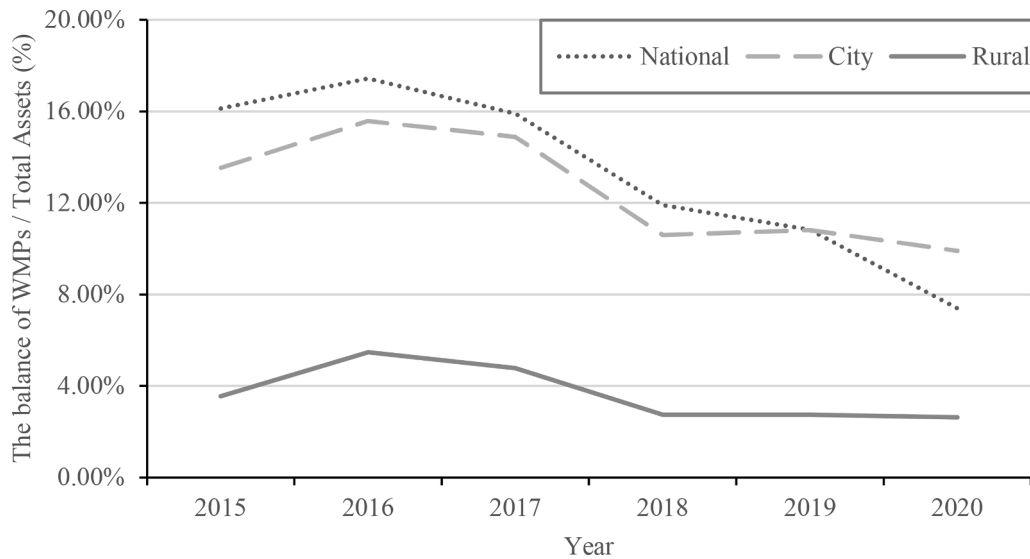


Fig. 1. The balance of WMPs over the assets of different types of commercial banks.

Note: National, city and rural refer to national commercial banks, city commercial banks and rural commercial banks, respectively. Source: China Banking Regulatory Commission (CBRC)

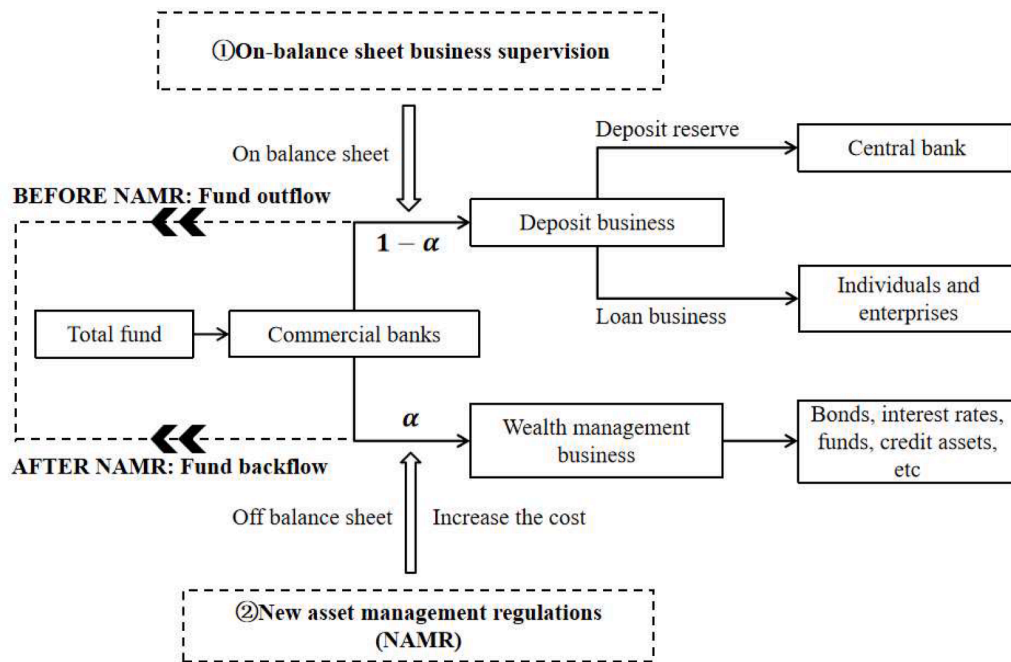


Fig. 2. Operation mode of commercial banks before and after NAMR implementation.

“Dodd-Frank Act” (U.S.) and “Securities and Futures Act” (Singapore), China’s NAMR is also designed to increase the transparency of financial products associated with shadow banking and to prescribe the responsibilities and functions of financial institutions and market regulators. In addition, NAMR has imposed strict regulations on financial products issued by financial institutions, which largely restricted commercial banks to participate shadow banking business.

Fig. 1 shows the balance of WMPs relative to the assets of different types of commercial banks. Since the new regulation was implemented, the balance of WMPs in all types of banks has declined. It can also be observed that national and city commercial banks have the largest ratios of WMPs in existence, and the impact of the new regulation on these two types of banks has lasted longer.

To understand the impact of NAMR on the participation of Chinese commercial banks in shadow banking, we have developed a model to depict the performance of commercial banks before and after the implementation of NAMR. Our model reveals that

commercial banks had a strong incentive to expand their shadow banking operations before NAMR. This incentive was significantly mitigated following NAMR, and we derived an optimal size for shadow banking from the banks' perspective. Moreover, we discuss how NAMR altered the risk and profitability of Chinese commercial banks based on micro-level evidence. Our empirical results indicate that the implementation of NAMR changed the issuance structure of WMPs, leading to a simultaneous reduction in banks' profits and risks. Notably, when the scale of WMPs decreased, the funds did not flow back to the banks' balance sheets.

Our work contributes in the following ways. First, it is one of a few studies that evaluate the impact of financial regulation on the shadow banking business as well as bank risk and performance in China. Prior researches primarily explore the development and operation mode, the motivation for the rise of bank wealth management (Acharya et al., 2021; Allen et al., 2019; Chen et al., 2018). Our findings extend the prior research by providing micro-level evidence on how the new regulations have changed the risks and profitability of commercial banks.

Second, we model the behaviors of banks and explore their incentive to issue WMPs. We find that banks tend to channel capital from saving accounts to WMPs to evade regulations on on-balance sheet. Money raised by WMPs are invested in risky assets to pursue higher profit. Our model also suggests an optimal scale of WMPs for banks after NAMR.

Finally, our study provides evidence that capital flowing out from on-balance sheet did not flow back and expected return of WMPs decreased after the new regulation. The finding demonstrates a more comprehensive evaluation of this new regulations.

2. Theoretical analysis

This section builds a theoretical analysis framework to examine how the implementation of NAMR affects the scale and profit of commercial banks' wealth management business. Fig. 2 clearly illustrates this theoretical framework. There are two business operated by banks, loan business and wealth management business. Loan business allows banks to provide funds to borrowers and charge interest. A contractual interest rate is charged in the loan business. Loan business is under the regulation of capital requirement and capital adequacy ratio (CAR) need to be fulfilled. Riskier loans are assigned higher risk weights, requiring commercial banks to provide more capital reserves and, therefore, banks choose their loan recipients carefully (Tucker and Lean, 2003). In contrast, wealth management business allows commercial banks to invest clients' funds and charge performance-based compensation, management, and service fees. Banks are not subject to CAR when they convert on-balance-sheet deposits into off-balance-sheet WMPs. Therefore, WMPs could be invested in projects or industries with high risk.

In our model, we assume that the economy has a formal banking system and a shadow banking system. The deposits (D) and equity (E) in formal banking system are divided into two parts—a central bank reserve (R) that cannot be used for profit, and a loan (L) that can be used for profit. Therefore, we assume that commercial banks have both assets, in the form of deposit reserves and loans:

$$L + R = D + E \quad (1)$$

where the reserve for deposits can be expressed as the deposit reserve ratio multiplied by the total amount of deposits taken by the bank.

$$R = \rho D \quad (2)$$

In our model, we assume our financial market contains two sectors: formal and informal banking system. S represents the amount of total fund in the financial market. A proportion of S , which is α will enter the informal banking system, while the remaining portion is channeled into the formal banking system. Thus, the total amount of wealth management products issued by commercial banks is:

$$W = \alpha S \quad (3)$$

The total wealth of the bank's deposit operations is:

$$D = (1 - \alpha)S \quad (4)$$

Before NAMR, the scale of WMPs issued by commercial banks was not restricted, due to the absence of relevant policies. Therefore, we assume that the marginal cost of WMPs issued remains unchanged—i.e., the cost C_1 is a linear function of the scale of issuance:

$$C_1 = \theta W = \theta \alpha S \quad (5)$$

where θ is the linear coefficient of the scale of wealth management issued.

The average revenue of commercial banks' wealth management business is:

$$I_w = mW = m\alpha S \quad (6)$$

where m is the average profit rate of wealth management business.

Based on Wolfe (2000), we construct the profit function of commercial banks. We denote the profit of commercial banks before NAMR implementation as π_1 , which contains on-balance sheet profit π_{b1} and off-balance sheet profit π_{o1} . On-balance sheet profit is loan interest income minus loan losses and deposit interest, and off-balance sheet profit is wealth management business income minus wealth management business costs.

$$\begin{aligned}\pi_{b1} &= (r_L - \phi)L - r_D D \\ &= [(r_L - \phi)(1 - \rho) - r_D](1 - \alpha)S + (r_L - \phi)E \\ &= \beta(1 - \alpha)S + (r_L - \phi)E\end{aligned}\quad (7)$$

$$\begin{aligned}\pi_{o1} &= I_{w1} - C_1 \\ &= (m - \theta)\alpha S\end{aligned}\quad (8)$$

where $\beta = (r_L - \phi)(1 - \rho) - r_D$, and $(m - \theta) - \beta > 0$.

Thus, the total profit of commercial banks is:

$$\pi_1 = \pi_{b1} + \pi_{o1} = \beta S + [(m - \theta) - \beta]\alpha S + (r_L - \phi)E \quad (9)$$

To maximize profit (π_1), banks increase α to the maximum value—i.e., $\alpha = 1$ —and all capital flows from the on-balance sheet deposit business to the off-balance sheet wealth management business.

After NAMR, we posit a quadratic relationship between costs and scale, implying that with an increase in the scale of WMPs, costs may exhibit accelerated growth. The NAMR prohibit financial institutions to operate multiple WMPs in a mixed asset pool. That is to say, each WMP has to be managed, accounted for independently, and each WMP should be invested in projects with same maturity. Therefore, when more financial products are issued, the more difficult it is to find suitable projects. Therefore, we assume that the cost of issuing WMPs by banks is exponentially related to the business size, which implies that the increase in cost may show an accelerated growth as the size of WMPs increases.

If the quadratic term coefficient is δ , then the cost function is:

$$C_2 = \theta W + \delta W^2 = \theta \alpha S + \delta \alpha^2 S^2 \quad (10)$$

Commercial banks are supposed to maximize profit π_2 by choosing the optimal α ; thus:

$$\begin{aligned}\pi_{b2} &= (r_L - \phi)L - r_D D \\ &= [(r_L - \phi)(1 - \rho) - r_D](1 - \alpha)S + (r_L - \phi)E \\ &= \beta(1 - \alpha)S + (r_L - \phi)E\end{aligned}\quad (11)$$

$$\pi_{o2} = I_w - C_2 = (m - \theta)\alpha S - \delta \alpha^2 S^2 \quad (12)$$

The total profit is:

$$\pi_2 = \pi_{b2} + \pi_{o2} = \beta S + [(m - \theta) - \beta]\alpha S - \delta \alpha^2 S^2 + (r_L - \phi)E \quad (13)$$

The optimal solution α^* can be derived from the first-order condition of profit maximization α :

$$\alpha^* = \frac{(m - \theta) - \beta}{2\delta S} \quad (14)$$

After NAMR, the operation of wealth management business is subject to many restrictions; thus, banks will not blindly engage in wealth management business, as there is an optimal solution for its scale, and the scale of WMPs will account for a smaller proportion of banks' available funds than before NAMR was enacted. If the change in the scale of banks' available funds is small, the implementation of the new regulations will reduce the absolute scale of WMPs issued by commercial banks.

3. Methodology

3.1. Data

Our bank-level data is collected from the banks' semi-annual financial reports in the Wind database¹ for the period 2015–2020. Data on WMP issuance is collected from the China Wealth Website,² which provides the main structural characteristics of WMPs. By matching the two datasets, we finally obtained 1538 bank-level samples from 261 commercial banks.

3.2. Variables and model

We select two indicators to measure the issuance structure of WMPs: the proportion of non-principal-guaranteed products (NPG) and the proportion of short-term wealth management products (ST). Meanwhile, Risk-Weighted Assets (RWA) and Return on Equity (ROE) are used to study the impact of NAMR on the risk and profitability of commercial banks.

¹ Wind (Wind Information Technology Co., Ltd.) is a leading financial software service company in mainland China, which has built a large-scale financial engineering and financial data warehouse with complete and accurate financial and securities data as its core in China. Wind's data covers stocks, bonds, funds, foreign exchange, financial derivatives, commodities, macroeconomics, financial news and other areas.

² China Wealth Website is the national banking industry's wealth management product inquiry website designated by the China Banking and Insurance Regulatory Commission; its website is found at <https://www.chinawealth.com.cn/zzlc/index.shtml>.

Table 1
Variables and explanation.

Category	Symbol	Description
Explanatory Variable	<i>NAMR</i>	The value of <i>NAMR</i> is 1 for the implementation period and all subsequent periods, and 0 for all previous periods
Dependent Variables	<i>LnWMP</i>	Ln (Number of WMPs issued by banks in the current period + 1)
	<i>NPG</i>	Proportion of the number of non-principal-protected WMPs issued in the current period to the number of all WMPs issued in the same period
	<i>ST</i>	Number of short-term WMPs issued in the current period as a proportion of the number of all WMPs issued in the same period
Control Variables	<i>RWAR</i>	Risk-weighted assets / Total assets
	<i>CAD</i>	Current assets / Current liabilities
	<i>ROE</i>	Net income / Equity
	<i>Size</i>	Ln (Commercial bank's asset size + 1)
	<i>LEV</i>	Total assets / Total liabilities
	<i>CIR</i>	Incremental business and management expenses for the period / Incremental operating revenues for the period
	<i>LDR</i>	Total loans / Total deposits
	<i>NII</i>	Non-interest income / Operating income for the period
	<i>Listed</i>	Listed banks = 1, non-listed banks = 0
	<i>GDP</i>	Province's semi-annual GDP cumulative year-on-year growth rate

Table 2
Descriptive statistics.

Variable	Full sample					Pre-NAMR		Post-NAMR	
	<i>N</i>	Mean	Std	Min	Max	<i>N</i>	Mean	<i>N</i>	Mean
<i>NAMR</i>	1538	0.632	0.482	0.000	1.000	566	0.000	972	1.000
<i>LnWMP</i>	1486	4.659	1.495	0.000	8.898	557	4.995	929	4.457
<i>NPG</i>	1451	0.709	0.311	0.000	1.000	556	0.662	895	0.738
<i>ST</i>	1453	0.233	0.215	0.000	1.000	557	0.364	896	0.151
<i>RWA</i>	1364	0.675	0.088	0.454	0.903	510	0.660	854	0.684
<i>ROE</i>	1538	0.053	0.031	−0.049	0.189	566	0.063	972	0.048
<i>Size</i>	1529	7.580	1.691	4.389	12.717	561	7.768	968	7.471
<i>NII</i>	1505	0.260	0.260	−0.580	1.013	558	0.260	947	0.259
<i>Listed</i>	1538	0.334	0.472	0.000	1.000	566	0.399	972	0.296
<i>LEV</i>	1529	1.081	0.016	1.047	1.145	561	1.075	968	1.084
<i>CIR</i>	1524	0.647	0.169	0.233	1.277	564	0.610	960	0.668
<i>LDR</i>	1511	0.681	0.117	0.407	1.049	559	0.633	952	0.709
<i>GDP</i>	1538	105.785	2.809	96.100	110.500	566	107.210	972	104.855

We constructed explanatory variables and control variables as Table 1 shows. To mitigate the endogeneity problem, all control variables were lagged by one period.

3.3. Descriptive statistics

Table 2 presents the descriptive statistics of key variables and compares them across pre-NAMR and post-NAMR.

3.4. Model

We used the event study method to set up a model investigating the impact of NAMR on the structure of WMPs issuance and on banks' risks and returns. In line with the study by Zhang et al. (2020), the model is set up as follows:

$$Y_{it} = \alpha_0 + \alpha_1 NAMR + \sum \delta X_{it-1} + \varepsilon_{it} \quad (15)$$

where Y_{it} is a set of dependent variables, including $LnWMP_{it}$, NPG_{it} , and ST_{it} ; X_{it} is the combination of control variables; and ε_{it} is random error term. The implementation date of NAMR is April 27, 2018, the first half of 2018 is set as the time point of the event.

4. Empirical results

4.1. Baseline regression

Table 3 presents the impact of NAMR on commercial banks' issuance of WMPs and its heterogeneous effects according to type of banks and level of capital adequacy ratio (CAR).

Our results indicate that the implementation of NAMR has demotivated commercial banks to issue WMPs, particular in city commercial banks. Meanwhile, it is interesting to find that banks with lower CAR levels suffered more severe impacts from NAMR. A

Table 3

The impact of NAMR on the issuance of wealth management products.

	(1) Full sample	(2) Full sample	(3) National	(4) City	(5) Rural	(6) High CAR	(7) Low CAR
Panel A	$LnWMP_{it}$						
NAMR	−0.419** (−2.44)	−0.997** (−2.54)	−1.200* (−1.82)	−1.041** (−2.07)	−0.551 (−0.73)	0.254 (0.33)	−1.102** (−2.27)
Control Variables	N	Y	Y	Y	Y	Y	Y
Time FE (half-year)	Y	Y	Y	Y	Y	Y	Y
N	2199	1299	170	586	533	474	746
adj. R ²	0.015	0.457	0.182	0.252	0.263	0.565	0.345
Panel B	NPG_{it}						
NAMR	0.209*** (5.64)	0.421*** (4.41)	0.242 (1.49)	0.437*** (3.93)	0.261 (1.13)	0.352 (1.33)	0.369*** (3.33)
Control Variables	N	Y	Y	Y	Y	Y	Y
Time FE (half-year)	Y	Y	Y	Y	Y	Y	Y
N	2265	1333	192	597	531	498	753
adj. R ²	0.057	0.155	0.035	0.247	0.164	0.162	0.116
Panel C	ST_{it}						
NAMR	−0.408*** (−15.38)	−0.324*** (−5.84)	−0.412*** (−3.71)	−0.257*** (−4.36)	−0.337** (−2.42)	−0.275** (−2.38)	−0.350*** (−5.14)
Control Variables	N	Y	Y	Y	Y	Y	Y
Time FE (half-year)	Y	Y	Y	Y	Y	Y	Y
N	2275	1335	193	597	532	499	754
adj. R ²	0.283	0.302	0.262	0.332	0.301	0.298	0.282

Note:.

(i) *, ** and *** means the significant level is less than 10 %, 5 % and 1 %, respectively.

(ii) Half-year next to FE means the time unit of sample used for estimation is a half year. T-statistics are reported in parentheses.

Table 4

The impact of NAMR on the comprehensive risk and return levels of commercial banks.

	(1) Full sample	(2) Full sample	(3) National	(4) City	(5) Rural	(6) High CAR	(7) Low CAR
Panel A	$RWAR_{it}$						
NAMR	−0.005 (−0.45)	−0.118*** (−4.35)	−0.044* (−1.86)	−0.105*** (−2.99)	−0.120*** (−2.91)	−0.096 (−1.60)	−0.110*** (−4.15)
Control Variables	N	Y	Y	Y	Y	Y	Y
Time FE (half-year)	Y	Y	Y	Y	Y	Y	Y
N	2059	1698	193	654	816	684	878
adj. R ²	0.003	0.263	0.547	0.379	0.369	0.242	0.474
Panel B	ROE_{it}						
NAMR	−0.060*** (−13.72)	−0.033*** (−4.26)	−0.035*** (−6.40)	−0.042*** (−5.20)	−0.012 (−0.78)	−0.006 (−0.32)	−0.042*** (−5.06)
Control Variables	N	Y	Y	Y	Y	Y	Y
Time FE (half-year)	Y	Y	Y	Y	Y	Y	Y
N	2516	1935	193	719	981	812	963
adj. R ²	0.160	0.236	0.618	0.347	0.191	0.209	0.313

Note:.

(i) *, ** and *** means the significant level is less than 10 %, 5 % and 1 %, respectively.

(ii) Half-year next to FE means the time unit of sample used for estimation is a half year. T-statistics are reported in parentheses.

possible reason for this is that banks with low levels of CAR had greater incentive to develop a shadow banking business, particularly off-balance sheet WMPs, to engage in regulatory arbitrage before NAMR's implementation; thus, they are more affected by the regulation.

We further explored the impact of NAMR on the risk and return levels of commercial banks. Column (2) in Table 4 shows that the regulation has been successful in reducing banks' off-balance sheet risk brought by their wealth management business. Meanwhile, the regulatory arbitrage window is closing for banks, and banks' profits are declining. In terms of RWAR, the coefficients of city and rural commercial banks are significant at the 1 % level with a greater degree of risk reduction, and the coefficient of national commercial banks is negative at the 10 % level.

The results in Panel B from column (3)–(5) indicate that ROE has declined for both national and city commercial banks, as they saw their earnings levels decline, while those of rural commercial banks did not change significantly.

Table 5

The impact of NAMR on the expected yield of wealth management products.

	(1) Full sample ER_{it}	(2) Full sample	(3) National	(4) City	(5) Rural	(6) High CAR	(7) Low CAR
NAMR	−1.509*** (−17.23)	−1.026*** (−6.73)	−1.254*** (−3.61)	−1.223*** (−4.83)	−1.163*** (−3.21)	−1.489*** (−3.36)	−0.943*** (−6.23)
MP	N	0.001*** (5.64)	0.001*** (3.09)	0.001*** (7.89)	0.001*** (12.07)	0.001*** (7.37)	0.001*** (4.59)
RT	N	−0.675*** (−9.93)	−0.523*** (−4.56)	−0.870*** (−17.97)	−0.719*** (−12.26)	−0.857*** (−19.19)	−0.643*** (−8.57)
Control Variables	N	Y	Y	Y	Y	Y	Y
Time FE (half-year)	Y	Y	Y	Y	Y	Y	Y
Bank FE	Y	Y	Y	Y	Y	Y	Y
N	429,915	354,046	192,128	120,860	40,058	63,714	290,332
adj. R^2	0.379	0.502	0.398	0.738	0.706	0.683	0.475

Note:.

(i) *, ** and *** means the significant level is less than 10 %, 5 % and 1 %, respectively.

(ii) Half-year next to FE means the time unit of sample used for estimation is a half year. T -statistics are reported in parentheses.**Table 6**

Growth of deposit of commercial banks after the implementation of NAMR.

	(1) Full sample DGR_{it}	(2) Full sample	(3) National	(4) City	(5) Rural	(6) High CAR	(7) Low CAR
NAMR	−0.061*** (−5.14)	−0.063*** (−3.29)	0.066*** (3.40)	−0.062* (−1.97)	−0.065** (−2.08)	−0.051* (−1.84)	−0.068*** (−2.73)
Control Variables	N	Y	Y	Y	Y	Y	Y
Time FE (half-year)	Y	Y	Y	Y	Y	Y	Y
N	2150	1914	193	708	973	801	956
adj. R^2	0.067	0.063	0.437	0.076	0.073	0.043	0.093

Note:.

(i) *, ** and *** means the significant level is less than 10 %, 5 % and 1 %, respectively.

(ii) Half-year next to FE means the time unit of sample used for estimation is a half year. T -statistics are reported in parentheses.

Columns (6) and (7) present the results of two subsamples grouped by CAR level. Panel A reveals that the risk level of low-CAR banks is significantly lower. This may be because low-CAR banks are already approaching the regulatory floor in the fierce competition for deposits. Increasing on-balance sheet lending will increase risk-weighted assets and further reduce capital adequacy, so these banks rely more on shadow banking to release pressure on indicators. Panel B reveals that low-CAR banks have significantly lower profitability; this indicates that the increased regulatory pressure inhibits the regulatory arbitrage of commercial banks, leading to a decrease in their return levels after NAMR.

4.2. The expected return of WMPs

We also explored the impact of NAMR on the expected return of individual financial products at the micro level by estimating model (16) on WMP-level dataset:

$$ER_{jt} = \beta_0 + \beta_1 NAMR_t + \beta_2 MP_j + \beta_3 RT_j + \sum \delta Z_{jt} + \rho_j + \mu_t + \varepsilon_{jt} \quad (16)$$

where ER_{jt} represents the expected return of the product at the time of issuance, and product-level features include the management period (MP_j) and revenue type of the product (RT_j), which is 1 for capital preservation and 0 otherwise. Z_{jt} contain the same bank-level characteristics of the product as before. In the end, we obtained 354,046 effective samples.

The regression results for the full sample are shown in Column (2) of Table 5, where the implementation of NAMR had a negative and significant effect on the expected return of products in general. The results indicate that after NAMR, the arbitrage opportunity brought by the operation modes of rolling issuance and borrowing short-term to invest long-term disappeared. Furthermore, the expected product yields are significantly lower in all groups of the sample, regardless of the bank type or CAR level.

4.3. Further discussion

Model (17) was developed to explore whether NAMR constrained the development of shadow banking in China, checking whether

Table 7
Robustness tests.

	(1) <i>LnWMP_{it}</i>	(2) <i>NPG_{it}</i>	(3) <i>ST_{it}</i>	(5) <i>RWA_{it}</i>	(6) <i>ROE_{it}</i>	(7) <i>DGR_{it}</i>
Panel A MLE estimation						
<i>NAMR</i>	−0.997** (−2.56)	0.433*** (4.28)	−0.308*** (−5.56)	−0.110*** (−4.00)	−0.030*** (−4.19)	−0.063*** (−3.02)
Control Variables	Y	Y	Y	Y	Y	Y
Time FE (half-year)	Y	Y	Y	Y	Y	Y
<i>N</i>	1299	1246	1248	1365	1538	1521
Log pseudo-likelihood	−2020.439	−166.713	405.472	1600.609	3441.220	1816.560
Panel B Bootstrap						
<i>NAMR</i>	−0.996*** (−3.69)	0.433*** (6.63)	−0.308*** (−6.85)	−0.110*** (−6.02)	−0.030*** (−5.32)	−0.063*** (−3.48)
Control Variables	Y	Y	Y	Y	Y	Y
Time FE (half-year)	Y	Y	Y	Y	Y	Y
<i>N</i>	1299	1246	1248	1365	1538	1521
adj. <i>R</i> ²	0.456	0.168	0.329	0.263	0.282	0.064
Panel C Annual Panel						
<i>NAMR</i>	−0.866*** (−6.81)	0.177*** (5.57)	−0.240*** (−11.53)	−0.030*** (−3.29)	−0.024*** (−6.45)	−0.001 (−0.12)
Control Variables	Y	Y	Y	Y	Y	Y
Time FE (half-year)	Y	Y	Y	Y	Y	Y
<i>N</i>	1228	1188	1191	1464	1677	1667
adj. <i>R</i> ²	0.456	0.191	0.252	0.203	0.451	0.059
Panel D Controlling the Impact of Other Regulation						
<i>NAMR</i>	−1.131** (−2.44)	0.419*** (4.36)	−0.337*** (−5.02)	−0.107*** (−3.36)	−0.039*** (−4.85)	−0.073*** (−2.86)
<i>Liquidity Ratio</i>	−0.001 (−0.46)	0.001 (1.19)	0.000 (0.52)	−0.000 (−0.70)	0.000 (0.72)	0.000** (2.55)
Control Variables	Y	Y	Y	Y	Y	Y
Time FE (half-year)	Y	Y	Y	Y	Y	Y
<i>N</i>	722	746	747	959	1036	1034
adj. <i>R</i> ²	0.468	0.136	0.286	0.225	0.271	0.084

Note..

(i) *, ** and *** means the significant level is less than 10 %, 5 % and 1 %, respectively.

(ii) Half-year next to FE means the time unit of sample used for estimation is a half year. *T*-statistics are reported in parentheses.

capital has flown back from the off-balance sheet business to the on-balance sheet business in commercial banks. We introduce growth rate of bank deposits *DGR_{it}* to represent the development of banks' deposit business.

$$DGR_{it} = \gamma_0 + \gamma_1 NAMR_t + \sum \delta X_{it-1} + \mu_t + \varepsilon_{it} \quad (17)$$

The results are shown in Table 6. In column (2), it is clear that *NAMR* has a negative and significant effect on the deposit growth rate, indicating that the new regulation did not bring back the capital flowing out from the on-balance sheet business of commercial banks. This implies that even though the regulation has curbed the flow of capital into off-balance sheet the capital flowing out from the on-balance sheet did not flow back and may have gone to “other areas” instead.

We conducted further heterogeneous analysis by studying different groups of banks, and we found that national commercial banks' growth rate of deposits is increasing, while that of city and rural commercial banks is decreasing. After grouping the banks according to the level of CAR, we found that banks with both high and low CAR do not achieve capital repatriation.

4.4. Robustness tests

We performed four robustness tests in our study. (i) Estimation using the maximum likelihood (MLE). (ii) Estimation using bootstrap repeated sampling. (iii) Estimation with Annual panel. (iv) Controlling for the impact of another policy. Another policy, the *Liquidity Risk Management Measures for Commercial Banks*,³ could have had an impact on our results. We add the *Liquidity Ratio* to the control variables, the results remain consistent and prove the robustness of the findings. Table 7

5. Conclusion

Our paper investigated the impact of *NAMR* on Chinese commercial banks. We found *NAMR* significantly changed the issuance

³ the *Liquidity Risk Management Measures for Commercial Banks* was implemented on May 23, 2018, establishing the supervisory indicators for commercial banks' liquidity risk and divided commercial banks into two categories according to whether their assets exceed 200 billion RMB. Both types of banks must examine the public indicators of liquidity ratio and liquidity matching ratio.

structure of WMPs, especially for city commercial banks and low-CAR banks. Meanwhile, NAMR reduced the risk and return level of commercial banks. Furthermore, our results suggest that the banking industry did not achieve the return of capital from on-balance sheet business after NAMR, and the implementation of NAMR also lower expected returns on WMPs.

Our findings generate significant implications for policymakers and market participants in China. On the one hand, the expansion of unregulated WMPs became a potential threat to the financial market and raised systemic risk; on the other hand, WMPs became a complement to SME financing, providing an impetus for their development. In this case, the purpose of regulating WMPs is no longer to eliminate them but to allow them to develop in a regulated environment and make their risks more manageable. Furthermore, our results suggest that the capital released from WMPs has not returned to on-balance sheet deposits. Therefore, a more comprehensive assessment of policy should be conducted to analyze the flow of capital after NAMR, and take further measures to avoid capital flows to other high-risk sectors. Finally, after the tightening of shadow banking activities, it is important to pay attention to the financial needs of SMEs and households.

CRedit authorship contribution statement

Yan Dong: Writing – review & editing, Validation, Supervision, Investigation, Conceptualization. **Mengyao Dong:** Writing – review & editing, Writing – original draft, Visualization, Software, Resources. **Suhang Tan:** Writing – review & editing, Software, Resources, Data curation. **Ran Ge:** Writing – review & editing.

Data availability

Data will be made available on request.

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